Anish Oturkar

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Education

DJ Sanghvi College of Engineering CGPA: 7.7
Vile Parle,Mumbai *Sep 22 - May 26*

Experience

Aule Space March 2026 – May 2026
Mechanical Intern
- Performed **Finite Element Analysis (FEA)** on the docking probe mechanism.
- Performed **Multi-Body Dynamics (MBD)** simulations using **MSC ADAMS** to simulate satellite approach.
- Worked on propulsion subsystem requirements for correct thruster sizing.

DJS Impulse Jan 2023 – January 2026
Co-Founder, Manager & Propulsion Lead
- Managed a team of 20 students. Handling timelines, using **systems engineering** methods, handling paperwork & technical reports, Handling systems integration.
- Designed and built two solid-propellant rocket motors which made me apply rocket propulsion and design concepts, Both the motors flew on separate rocket missions
- Worked on the avionics, both on software (helped built the apogee detection logic) and hardware, PCB basics & circuits, used languages like **c++ and python** for embedded software and GUI.
- Simulated in-flight drag/lift forces and engine nozzle exit pressure/velocity in ANSYS Fluent to validate analytical propulsion calculations. Using **ANSYS static structural** for stress analysis of the solid rocket motor to validate the analytical calculations.

Eyantra Robotics competition/Internship - IIT Bombay Sep. 24 – Jun 2025
Team member
- robotics competition which focuses on warehouse bots like robotic arm,differential drive rover.
- Developing skills in ROS2 Simulations software like Gazebo, Rviz.
- understanding concepts like ROS framework, mapping,localization, robotic arm kinematics.
- Learning python scripting and logic and trying to implement the simulations to real physical world.

Projects

Self balancing bot — Build a small 2 wheel bot which balances itself. employed & Tuned a PID loop to make sure the bot remains steady.

Development of a roll detection sensor — final year project in which we built a mechanically triggered sensor which detects vehicle roll.

Courses

NPTEL - Rocket Propulsion — Completed March 2023
Manufacturing Guidelines for Product Design — Completed March 2025
Systems Engineering – Theory & Practice — Completed March 2026
Principle Of Hydraulic Machines & System Design — Ongoing

Skills

Computer-Aided Design (CAD), Finite element Analysis (FEA), Rocket propulsion, GD&T, Failure Mode and Effects Analysis (FMEA), Design for Manufacturing (DFM), Multi-Body Dynamics (MBD), Python, Systems Engineering.